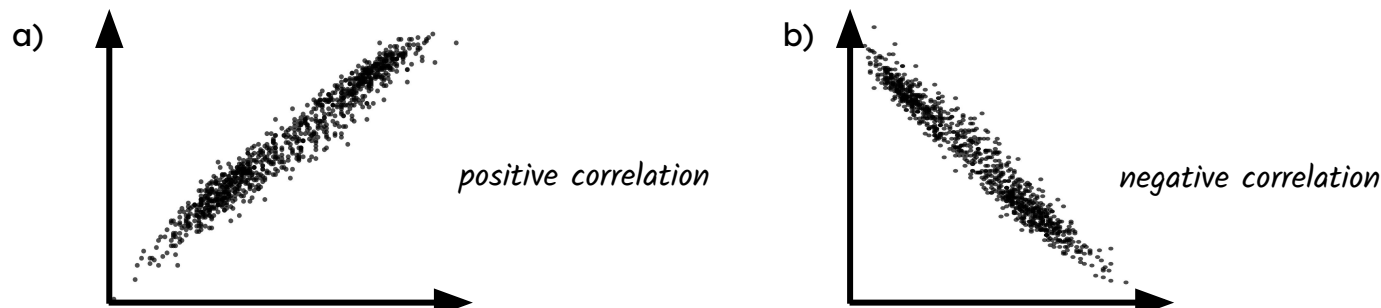




Correlation

Task A: Recognising correlation in bivariate data

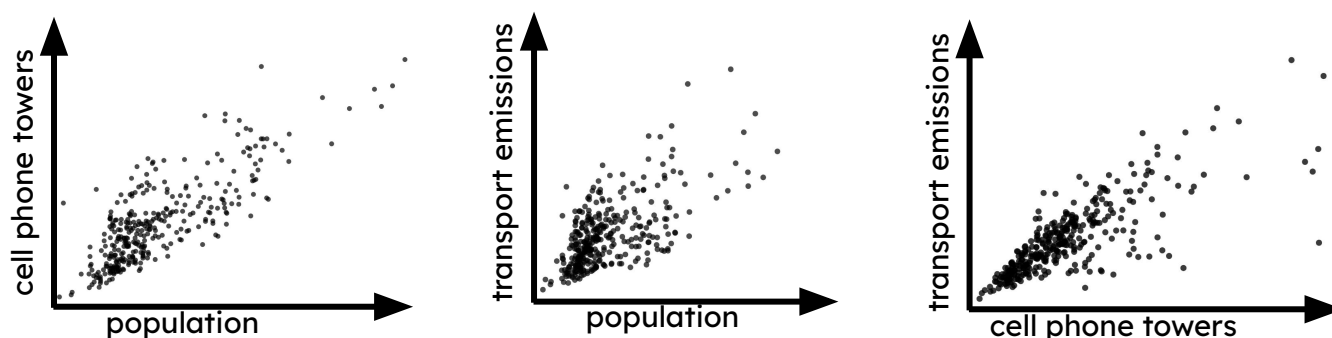
1) Fully describe the **correlation** in each scatter graph.



2) For each set of bivariate data, tick whether you would expect the **correlation** to be positive or negative.

	positive	negative
children's heights and their shoe sizes	✓	
steepness of a hill and walking speed up the hill		✓

3) The three scatter graphs show data from the Office for National Statistics. Each point represents a region in England and Wales in 2015.



Jacob thinks that the third scatter graph shows that cell phone towers cause more traffic pollution because there is a positive correlation.

Explain why Jacob could be wrong.

Correlation does not always mean that one variable is affecting the other.

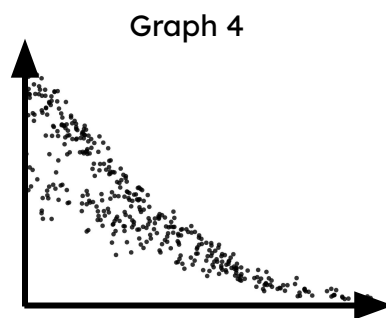
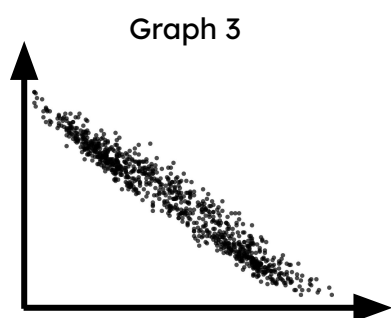
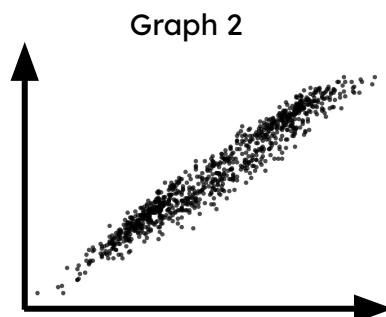
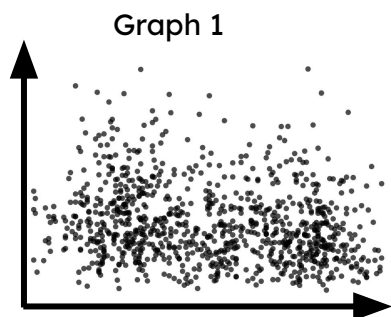
Correlation may suggest there is an association between the variables, not necessarily a cause and effect relationship.

The number of cell phone towers and the amount of traffic emissions could both be affected the population of the region.



Task B: Interpreting bivariate data without correlation

1) Here are four scatter graphs.



- a) Which graph shows a **positive correlation**? *Graph 2*
- b) Which graph shows a **negative correlation**? *Graph 3*
- c) Which graph shows an association which is not linear? *Graph 4*
- d) Which graph shows no association? *Graph 1*

